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BS Chemistry 3-A
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AI Tool Prompt Engineering Report

I. AI Tool Overview

A. AI Selected Tool – Gemini

B. Purpose/Domain

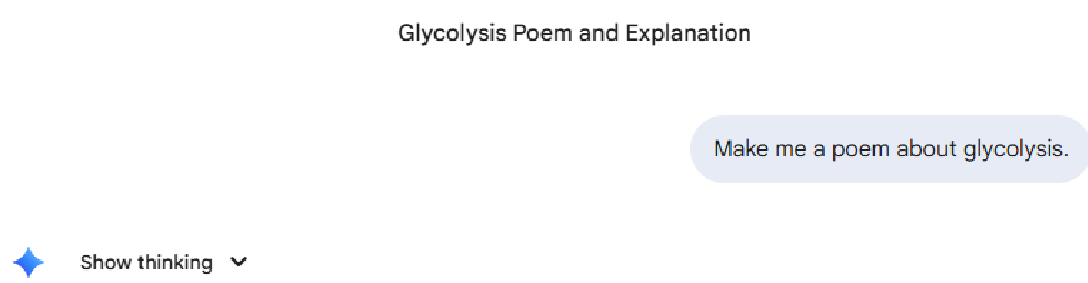
I will be using Gemini AI to create a poem about glycolysis. Initially, I will be writing a prompt like how people usually write prompts on AI. Afterwards, the poem generated by AI will be copied and it will be pasted to another chat but using the same AI tool, but I will be asking Gemini to generate an optimized prompt that can get me an enhanced response.

The optimized prompt generated by Gemini will be sent to the previous chat with the original prompt and compare whether the response from the original prompt was enhanced or not.

C. Reason behind the chosen AI tool

I chose Gemini AI because based on my previous experiences with AI tools like Chatgpt, Gemini AI produced better results compared to Chatgpt.

II. Original prompt



A. Original Prompt Output

Show thinking

The Cytoplasmic Shuffle: A Glycolysis Tale

In the quiet cytosol, away from the light,
A six-carbon sugar begins its long flight.
Glucose arrives, so stable and sweet,
But it needs a small spark to make it complete.

Two **ATP** enter, an investment of gold,
To prime the reaction and make the bond bold.
Hexokinase acts, the trap has been set,
Fructose-1,6-bisphosphate is the best bet.

Then comes the cleavage, a structural split,
Six carbons break down, bit by bit.
G3P rises, while DHAP transforms,
Two identical dancers in metabolic storms.

Now for the payoff, the energy flow,
NAD+ is reduced as the electrons go.
Four **ATP** rise from the substrate's embrace,

III. AI-Assisted Prompt

Optimizing Glycolysis Poem Prompt

Help me improve this prompt: "Make me a poem about glycolysis." I want the new prompt to include all enzymes involved in glycolysis.

That's a fun challenge! Glycolysis is essentially a ten-act biological drama, and while a basic poem is a good start, adding all ten enzymes turns it into a technical masterpiece.

To get the best result from an AI, you want to specify the **tone**, the **narrative flow**, and how the **enzymes** should be integrated (e.g., as characters or bolded terms).

The Optimized Prompt

> "Write a structured, 10-stanza poem that chronicles the pathway of glycolysis. Each stanza must focus on one specific chemical reaction, explicitly naming the substrate and the enzyme responsible for the transformation.
>
> **Please include these enzymes in chronological order:**
>
> 1. Hexokinase
> 2. Phosphoglucose isomerase
> 3. Phosphofructokinase-1
> 4. Aldolase
> 5. Triosephosphate isomerase
> 6. Glyceraldehyde 3-phosphate dehydrogenase
> 7. Phosphoglycerate kinase
> 8. Phosphoglycerate mutase
> 9. Enolase
> 10. Pyruvate kinase
>
> Use a rhythmic, educational tone (similar to a rhythmic 'slam poetry' style) and ensure the distinction between the energy investment phase and the energy payoff phase is clear."

A. Optimized Prompt Output

Glucose walks into the cytosol, stable and sweet,
But the cell needs it trapped, so it doesn't retreat.
Hexokinase strikes with a lightning-fast hit,
Taking **Glucose** and adding a phosphate to it.
One **ATP** spent, the investment is made,
Now **Glucose-6-phosphate** can't leave or evade.

Glucose-6-phosphate, you're a bit too square,
Time to rearrange the atoms you wear.
Phosphoglucose isomerase enters the scene,
Shifting the bonds of the hexose machine.
A five-membered ring is the brand new state,
Say hello to **Fructose-6-phosphate**.

Fructose-6-phosphate, the pace is now rising,
The cell's second payment is not so surprising.
Phosphofructokinase-1 takes the lead,
Satisfying the metabolic hunger and greed.
Another **ATP** sacrificed for the gate,
To create **Fructose-1,6-bisphosphate**.

Six carbons are heavy, six carbons are wide,
Time for **Aldolase** to choose a side.
It cuts the big sugar right down the core,
Leaving two three-carbon pieces on the floor.
DHAP and **G3P** are the twins of the split,
A symmetrical break that's a perfect fit.

DHAP, you're the twin that's a little bit slow,
But **Triosephosphate isomerase** is ready to go.
It flips your structure, it mirrors your face,
Until two **G3P** are ready to race.
The investment is over, the debt has been paid,
Now watch as the energy payoff is made!

Two **G3P** enter the oxidation storm,
Glyceraldehyde 3-phosphate dehydrogenase begins to transform.
NAD+ comes in, grabbing electrons and light,
While an inorganic phosphate joins the fight.
NADH is the trophy, the energy's great,
Behold the high-energy **1,3-Bisphosphoglycerate**.

1,3-Bisphosphoglycerate, you're holding the heat,
Phosphoglycerate kinase makes the transaction complete.
It strips off a phosphate, ADP takes the prize,
Our first **ATP** starts to rise.
We're back to the "net" as the molecules state,
The arrival of steady **3-Phosphoglycerate**.

3-Phosphoglycerate, move that group to the middle,
Phosphoglycerate mutase solves the chemical riddle.
From carbon number three to carbon number two,
A molecular migration for me and for you.
Stability shifts as we navigate the view,
Now we're holding **2-Phosphoglycerate** too.

2-Phosphoglycerate, drop a molecule of water,
Enolase acts as the dehydration spotter.
It creates a double bond, a high-energy zest,
Putting the chemical structure to the test.
The tension is building, the power is set,
Phosphoenolpyruvate is the strongest yet.

PEP, you're the end, the final big boom,
Pyruvate kinase has entered the room.
One last phosphate is handed to the ADP line,
Two more **ATP**! Everything's fine.
The sugar is broken, the energy's won,
Two **Pyruvates** remain—Glycolysis is done!

IV. Comparison and Analysis

The objective of this laboratory activity is to compare whether the response of an AI tool such as Gemini will be enhanced using an optimized prompt that was generated by the same Gemini tool, except that the writing of the prompt happened at a different chat. Based on the response results from the original prompt (II.A) and the optimized prompt (IIIA), the response from the optimized prompt was more detailed in generating the poem about glycolysis. Conversely, the response from the original prompt was quite lacking, for it did not cover all the enzymes that participate in glycolysis. The quality of response obtained was dependent on the specificity of the prompt given to Gemini.

The glycolytic process contains ten steps, and in each step, it requires an enzyme. In the original prompt, this was written simplistically—without providing further contexts and details on what should be seen in the output. As a result, Gemini left out a lot of details on glycolysis. On the other hand, when the optimized prompt was sent to Gemini, there was an enhancement in its response. The optimized prompt response accounted for all enzymes and even other molecules that are participating in glycolysis.

The comparison revealed significant differences between the two outputs, illustrating the dramatic impact of prompt quality. The output generated by the optimized prompt was unequivocally the more useful and accurate of the two, as it achieved the underlying goal of the exercise: a comprehensive, educational poem about glycolysis. The initial, unassisted prompt produced a poem that was highly conceptual and poetic but lacked the necessary scientific depth, failing to mention several critical enzymes required for the ten-step process. This lack of accuracy rendered the initial output unsuitable for an academic context, reinforcing the notion that simple, conversational prompts often result in superficial content.

The improved prompt dramatically affected the clarity, depth, and relevance of the output. Clarity was enhanced because the refined instructions explicitly detailed the desired structure and required components, removing ambiguity about the scope of the poem. Depth saw the most profound improvement; while the original output offered a general overview of the process, the optimized output delved into the specific enzymatic reactions, including molecules like ATP and NADH, making the poem a true summary of the biochemical pathway. Relevance was also maximized, as the optimized prompt ensured the resulting poem was directly applicable to a study of biochemistry, moving beyond generic creative writing.

The specific changes in the prompt, derived through assistance from Gemini itself, focused on increasing the contextual information and adding constraints. By explicitly instructing Gemini to include the sequence of ten steps, the role of each enzyme, and the specific reactants and products, the prompt moved from a general request ("write a poem about glycolysis") to a highly structured directive. This shift from simplicity to specificity is the key learning in prompt engineering from this activity. It demonstrates that treating the Large Language Model (LLM) not just as a creative partner but as a sophisticated tool requiring precise instructions is essential for obtaining high-quality, factual, and complete results. Therefore, from this laboratory activity, it is evident that the higher the specificity of the prompt delivered to Gemini, the better the probability of obtaining desirable results.